WORKED SOLUTION

STEP II 2023, Question 11

Probability and Statistics •

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 13.5, *Asking Awkward Questions*: https://beyondstep.keithmcnulty.org/ch13/awkward-questions.html

The key idea

Y is a *mixture*: first pick a distribution at random, then draw from it. Its mean is the weighted average of the two means, but its variance has an extra term, because the spread between the two means adds variability of its own. Parts (ii) and (iii) both toss a coin before using a die, yet they behave completely differently. In (ii) a fresh coin is tossed for every throw, so the mixing averages out. In (iii) a single coin decides everything, and that extra term ends up dominating.

Solution

Part (i)

Choosing which variable to use is done independently of the values of X_1 and X_2, so

$$\mathrm{P}(Y = x_i) = p\,a_i + q\,b_i \qquad (i = 1, \ldots, n).$$

Hence

$$\mathrm{E}(Y) = \sum_i x_i(pa_i + qb_i) = p\sum_i x_i a_i + q\sum_i x_i b_i = p\mu_1 + q\mu_2.$$

In the same way, $\mathrm{E}(Y^2) = p\,\mathrm{E}(X_1^2) + q\,\mathrm{E}(X_2^2) = p(\sigma_1^2 + \mu_1^2) + q(\sigma_2^2 + \mu_2^2)$. So

$$\begin{aligned}\mathrm{Var}(Y) &= p\sigma_1^2 + q\sigma_2^2 + p\mu_1^2 + q\mu_2^2 - (p\mu_1 + q\mu_2)^2 \\ &= p\sigma_1^2 + q\sigma_2^2 + p(1-p)\mu_1^2 + q(1-q)\mu_2^2 - 2pq\mu_1\mu_2 \\ &= p\sigma_1^2 + q\sigma_2^2 + pq\mu_1^2 + pq\mu_2^2 - 2pq\mu_1\mu_2 \\ &= p\sigma_1^2 + q\sigma_2^2 + pq(\mu_1 - \mu_2)^2,\end{aligned}$$

using $1 - p = q$ and $1 - q = p$. This is the required result.

Part (ii)

Let H be the event that the coin shows heads, and S the event that the die shows a six. Coin and die are independent, so

$$\mathrm{P}(B=1) = \mathrm{P}(H)\mathrm{P}(S) + \mathrm{P}(H^c)\mathrm{P}(S^c) = \tfrac{1}{2}\cdot\tfrac{1}{6} + \tfrac{1}{2}\cdot\tfrac{5}{6} = \tfrac{1}{2}.$$

So each B is 0 or 1 with probability $\frac{1}{2}$ each. Z_1 is the number of 1s among n independent copies of B, each a success with the same probability $\frac{1}{2}$. That is exactly the binomial set-up, so $Z_1 \sim \mathrm{B}(n, \frac{1}{2})$, with mean $\frac{n}{2}$ and variance $\frac{n}{4}$.

For large n we approximate Z_1 by $\mathrm{N}(\frac{n}{2}, \frac{n}{4})$, whose standard deviation is $\frac{\sqrt{n}}{2}$. Being within 10% of the mean means $|Z_1 - \frac{n}{2}| \le \frac{n}{20}$. Standardising, with $Z \sim \mathrm{N}(0,1)$,

$$\mathrm{P}\big(|Z_1 - \tfrac{n}{2}| \le \tfrac{n}{20}\big) \approx \mathrm{P}\Big(|Z| \le \frac{n/20}{\sqrt{n}/2}\Big) = \mathrm{P}\Big(|Z| \le \frac{\sqrt{n}}{10}\Big) = 2\Phi\Big(\frac{\sqrt{n}}{10}\Big) - 1.$$

As $n \to \infty$, $\sqrt{n}/10 \to \infty$, so $\Phi(\sqrt{n}/10) \to 1$ and the probability tends to $2 - 1 = 1$. (A continuity correction changes the bound by only $\frac{1}{\sqrt{n}}$ standard deviations and does not affect the limit.)

Part (iii)

Let X_1 be the number of sixes and X_2 the number of non-sixes among the n dice. Then $X_1 \sim \mathrm{B}(n, \frac{1}{6})$ and $X_2 \sim \mathrm{B}(n, \frac{5}{6})$. Now Z_2 takes the value of X_1 with probability $p = \frac{1}{2}$ and that of X_2 with probability $q = \frac{1}{2}$, and the coin is independent of the dice. This is exactly the situation of part (i), with

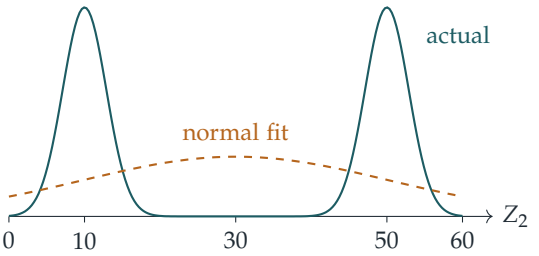
$$\mu_1 = \tfrac{n}{6}, \quad \mu_2 = \tfrac{5n}{6}, \quad \sigma_1^2 = \sigma_2^2 = n\cdot\tfrac{1}{6}\cdot\tfrac{5}{6} = \tfrac{5n}{36}.$$

So

$$\mathrm{E}(Z_2) = \tfrac{1}{2}\cdot\tfrac{n}{6} + \tfrac{1}{2}\cdot\tfrac{5n}{6} = \tfrac{n}{2},$$
$$\mathrm{Var}(Z_2) = \tfrac{1}{2}\cdot\tfrac{5n}{36} + \tfrac{1}{2}\cdot\tfrac{5n}{36} + \tfrac{1}{4}\Big(\tfrac{4n}{6}\Big)^2 = \frac{5n}{36} + \frac{n^2}{9}.$$

Ask yourself What does the distribution of Z_2 actually look like for large n?

With probability $\frac{1}{2}$, Z_2 is close to $\frac{n}{6}$, and with probability $\frac{1}{2}$ it is close to $\frac{5n}{6}$. Each of these humps has standard deviation $\frac{\sqrt{5n}}{6}$, which is tiny compared with their separation $\frac{2n}{3}$. So the distribution is *bimodal*: two narrow peaks, with almost no probability near the mean $\frac{n}{2}$. A normal distribution has a single peak at its mean, so it is exactly wrong here: it puts the most probability where Z_2 almost never lands. There is a second problem too. The standard deviation is about $\frac{n}{3}$, so $\mathrm{N}(\frac{n}{2}, \frac{n^2}{9})$ gives probability about $\mathrm{P}(Z < -1.5) \approx 0.067$ to negative values, and the same to values above n, though $0 \le Z_2 \le n$ always. The sketch shows the two pictures for $n = 60$.



Now for the final part. Z_2 is within 10% of its mean when $0.45n \le Z_2 \le 0.55n$. Conditioning on the coin,

$$\mathrm{P}(0.45n \le Z_2 \le 0.55n) = \tfrac{1}{2}\,\mathrm{P}(0.45n \le X_1 \le 0.55n) + \tfrac{1}{2}\,\mathrm{P}(0.45n \le X_2 \le 0.55n).$$

For the first term, $\mathrm{P}(0.45n \le X_1 \le 0.55n) \le \mathrm{P}(X_1 \ge 0.45n)$, and $0.45n - \frac{n}{6} = \frac{17n}{60}$. Using the normal approximation $X_1 \approx \mathrm{N}(\frac{n}{6}, \frac{5n}{36})$, whose standard deviation is $\frac{\sqrt{5n}}{6}$,

$$\mathrm{P}(X_1 \ge 0.45n) \approx \mathrm{P}\Big(Z \ge \frac{17n/60}{\sqrt{5n}/6}\Big) = 1 - \Phi\Big(\frac{17\sqrt{n}}{10\sqrt{5}}\Big) \longrightarrow 1 - 1 = 0.$$

The second term is the mirror image: $X_2 = n - X_1$, so $0.45n \le X_2 \le 0.55n$ exactly when $0.45n \le X_1 \le 0.55n$, and the second term equals the first. Hence the probability that Z_2 is within 10% of its mean is at most about $1 - \Phi\big(\frac{17\sqrt{n}}{10\sqrt{5}}\big)$, which tends to 0 as $n \to \infty$.

This step does not really depend on the approximation. Chebyshev's inequality gives exactly

$$\mathrm{P}\big(|X_1 - \tfrac{n}{6}| \ge \tfrac{17n}{60}\big) \le \frac{5n/36}{(17n/60)^2} = \frac{500}{289\,n} \longrightarrow 0.$$

Checks

- In (i), if $\mu_1 = \mu_2$ the extra term vanishes, and if $\sigma_1 = \sigma_2 = 0$ then Y takes the value μ_1 or μ_2 only, with variance $pq(\mu_1 - \mu_2)^2$, as for a scaled Bernoulli variable.
- In (ii), the exact binomial probability for $n = 400$ is 0.9598, against the normal value $2\Phi(2) - 1 = 0.9545$. For $n = 1000$ the two values are 0.9986 and 0.9984.
- In (iii), a computer simulation of Z_2 with $n = 30$ gave mean 15.00 and variance 104.2, against the formula's 15 and 104.17. The exact probability of being within 10% of the mean is about 3×10^{-11} when $n = 100$.

Common slips

- Giving $\mathrm{Var}(Y) = p\sigma_1^2 + q\sigma_2^2$, forgetting that the choice between the two means is itself a source of variation.
- Treating Z_2 in (iii) like Z_1 in (ii). In (ii) the coin is re-tossed for every die, but in (iii) one toss governs all n dice, so the outcomes are not independent.

- Explaining the poor fit only by "the variance is large". The real problem is the shape: two separate peaks, with very little probability near the mean.

Beyond STEP

Part (i) is the simplest case of the *law of total variance*, $\mathrm{Var}(Y) = \mathrm{E}\big(\mathrm{Var}(Y \mid C)\big) + \mathrm{Var}\big(\mathrm{E}(Y \mid C)\big)$, where C is the coin. Part (ii) is the weak law of large numbers, which a first course proves rigorously using Chebyshev's inequality. Part (iii) shows why that law needs independence: one shared coin is enough to destroy it. Mixture distributions like Z_2 are a staple of statistical modelling. Blitzstein and Hwang, *Introduction to Probability*, covers all three ideas with many examples.

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